Experiment 04

Title

Telecom Customer Churn Prediction using Machine Learning Models

SAP ID:	590013857
Dataset:	`https://www.kaggle.com/blastchar/telco-customer-churn`

AIM

To build and compare multiple machine learning models to predict telecom customer churn and identify key factors influencing customer retention.

THEORY

Customer churn prediction is a classification problem where the goal is to identify whether a customer will leave a service.

Key concepts:

- **Classification Models**: Logistic Regression, KNN, SVM, Decision Tree, Random Forest
- **Evaluation Metrics**: Accuracy, Precision, Recall, F1-score, AUC-ROC
- **ROC Curve**: Measures model performance across thresholds
- **Confusion Matrix**: Shows prediction correctness
- **Feature Importance**: Identifies key churn-driving factors

Commands/Code Used

```
# Core steps of Experiment 4

# Load dataset
df = pd.read_csv("Telco-Customer-Churn.csv")

# Preprocessing
df = df.drop_duplicates()

# Encoding + Scaling
# (Label encoding + StandardScaler applied)

# Train multiple models
models = [
    LogisticRegression(),
    KNeighborsClassifier(),
    SVC(probability=True),
```

```
    RandomForestClassifier(),
    DecisionTreeClassifier()
]

# Train and evaluate models
for model in models:
    model.fit(X_train, y_train)

# Evaluate using:
# Accuracy, Precision, Recall, F1, AUC-ROC

# Generate visualizations:
# - ROC curves
# - Confusion matrix
# - Feature importance
```

Results

1. Feature Importance (Random Forest)

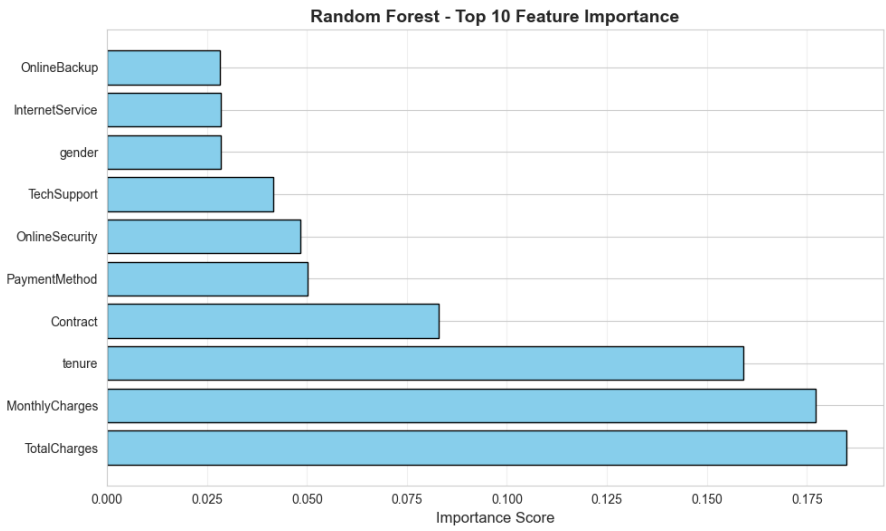

Figure 1: Top features influencing customer churn

2. Model Performance Comparison

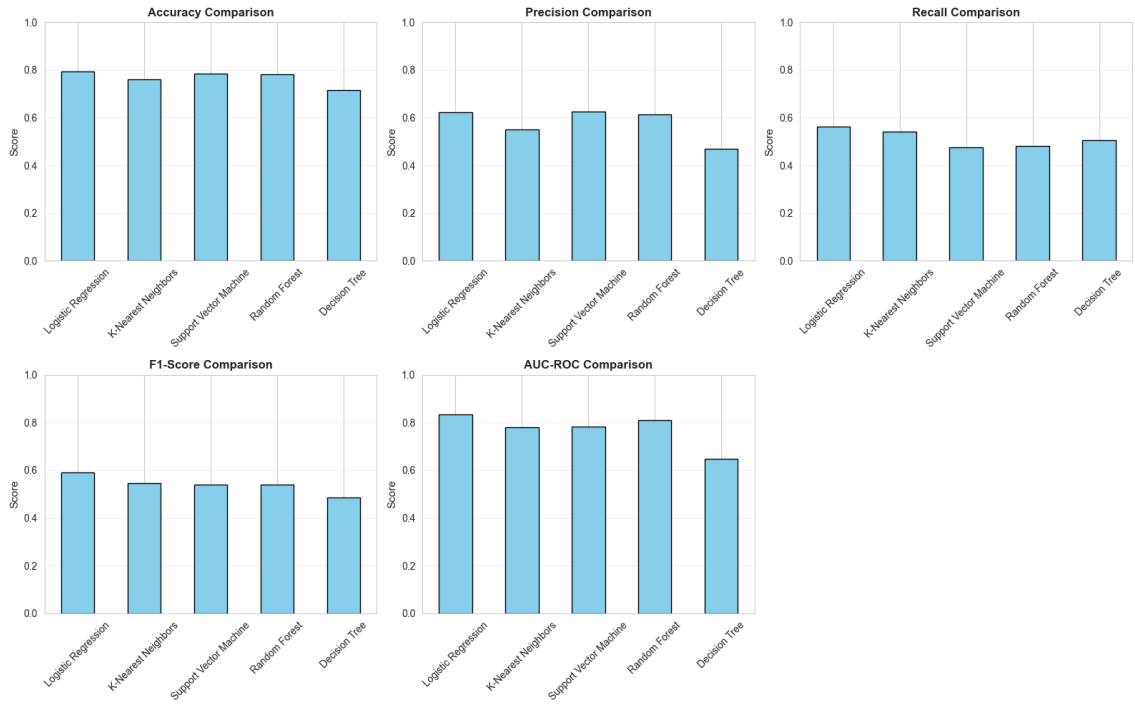

Figure 2: Accuracy, Precision, Recall, F1-score and AUC comparison

3. ROC Curve Comparison

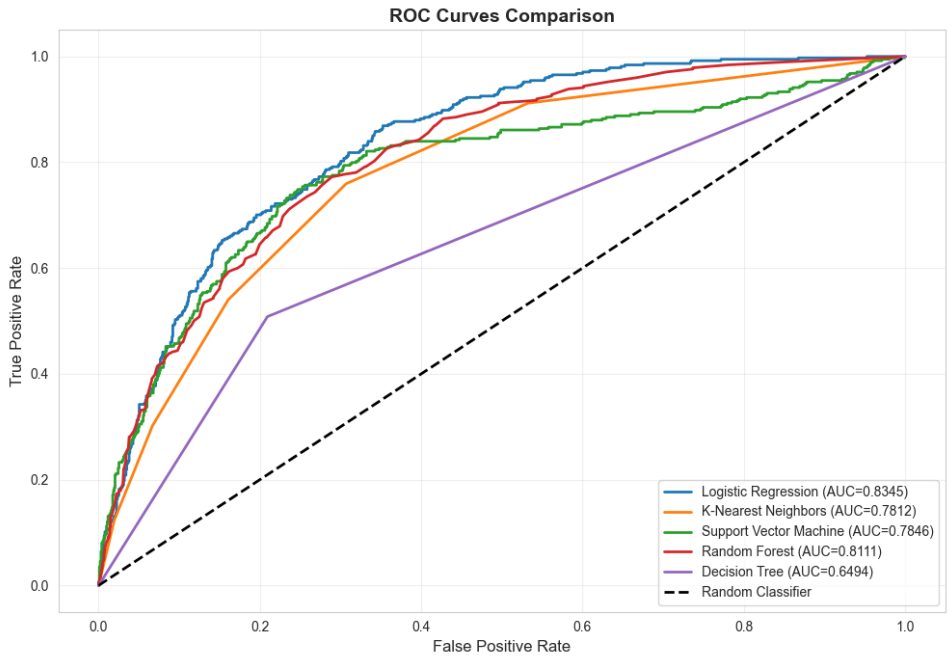

Figure 3: ROC curves for all models

4. Confusion Matrices

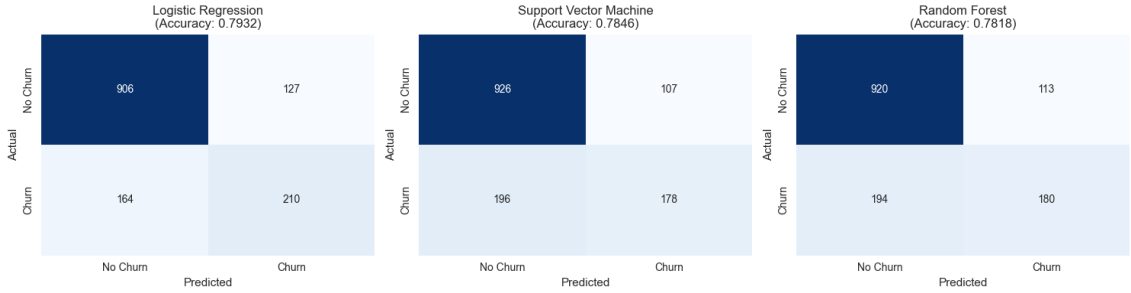

Figure 4: Confusion matrices of top performing models

Conclusion

In this experiment, multiple machine learning models were trained and evaluated for telecom churn prediction.

The comparison of models showed that ensemble methods like Random Forest perform well due to their ability to capture complex patterns. ROC curves and confusion matrices provided deeper insights into model performance.

Feature importance analysis revealed that factors such as contract type, tenure, and service-related features significantly influence churn.

Overall, this experiment demonstrates how multiple models and evaluation techniques can be used to build a robust predictive system for real-world applications.